Cassandra Engstrom

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Overview

My interests lie in mathematical descriptions of spatial learning, exploration, search and navigation, with attention paid to how these processes might be supported by the brain. I use behavioral studies in VR, computational modeling and agent simulations to address related questions.

Education

Brown University, Providence, RI **Sep 2022 – May 2027 (Expected)**
PhD, Cognitive Science, Advisor: Dr. William Warren Jr.

The Graduate Center, City University of New York, New York, NY **Sep 2019 – May 2022**
MSc, Cognitive Neuroscience, Advisor: Dr. Tony Ro

Harvard Extension School, Online **Jan 2019 – May 2019**
Nonmatriculating coursework

The Cooper Union, New York, NY **Sep 2011 – May 2016**
B.Arch (Bachelor of Architecture)

Grants and Awards

2025 –	T32 Training Program for Interactionist Cognitive Neuroscience (ICoN) (Brown University)
2020, 2021	Dean's University Fellowship (The Graduate Center, City University of New York)
2016	The American Institute of Architects Henry Adams Certificate of Merit (The Cooper Union)
2016	The Martin J. Water's Memorial Award (The Cooper Union)
2015	The William Cooper Mack Thesis Travel Fellowship (The Cooper Union)
2015	The Benjamin Menschel Fellowship Award (The Cooper Union)

Conference Proceedings

Engstrom, C. and Warren, WH (2026, July). How do humans search a large-scale space? Oral presentation at the *Interdisciplinary Navigation Symposium*, Banff, Alberta, Canada.

Engstrom, C. and Warren, WH (2026, May). The road ahead: knowledge does not replace vision in navigational decision-making. Poster presented at *Vision Sciences Society*, St. Pete Beach, FL, USA.

Engstrom, C. and Warren, WH (2025, August). A simple visual heuristic allows maze navigators to choose shorter routes under combinatorial uncertainty. Poster presented at *European Conference on Visual Perception*, Mainz, Germany.

Engstrom, C., Srinivasan, M. and Warren, WH (2025, July). Navigators trade off proximal and distal energy minimization strategies in uncertain environments. Poster presented at *XV Progress in Motor Control*, Kingston, RI, USA.

Engstrom, C. and Warren, WH (2024, June). Local navigation strategies guide global route selection. Poster presented at *International Conference in Perception and Action*, Trondheim, Norway.

Engstrom, C. and Warren, WH. (2024, June). Naive navigation strategies utilize local perceptual information and are modulated by destination proximity. Poster presented at *Interdisciplinary Navigation Symposium (iNAV)*, Merano, Italy.

Engstrom, C. and Warren, WH (2024, May). Global route selection using local visual information. Poster presented at *Vision Sciences Society*, St. Pete Beach, FL, USA.

Engstrom, C. and Colwill, R (2023, March). Interval scheduling of spatial context exposure impacts 10 different behavioral indices in larval zebrafish. Poster presented for *Eastern Psychological Association*, Boston, MA, USA.

Teaching

Spring 2026	Adjunct Instructor	Environmental Design Research (Roger Williams University, ARCH 522)
Spring 2025	Graduate Teaching Assistant	Making Decisions (Brown, CLPS 0220)
Fall 2024	Graduate Teaching Assistant	Animal Behavior (Brown, CLPS 0110)
Spring 2024	Graduate Teaching Assistant	Learning and Conditioning (Brown, CLPS 0100)
Fall 2023	Graduate Teaching Assistant	Brain Damage and the Mind (Brown, CLPS 0450)
Summer 2023	Graduate Teaching Assistant	Free Will and the Brain: Neuroeconomics and the Science of Decision-Making (Brown, CEBN 0937 01)
Summer 2023	Graduate Teaching Assistant	Data Skills for Bioinformatics (Brown, CECS 0931)

Summer 2022, 2023	Guest Lecturer	The Science of Psychology (Columbia University, S1001. Online)
Summer, Fall 2020	Adjunct Instructor	Physiological Psychology (Lehman College, PSY 316. Online)
Spring, Summer, Fall 2021	Adjunct Instructor	Physiological Psychology (Lehman College, PSY 316. Online)
Spring, Summer 2022	Adjunct Instructor	Physiological Psychology (Lehman College, PSY 316. Online)

Workshops and Training Courses

2026	Embodied Brain Technology Practicum, *Brown University*
2026	Brown Ethics and Responsible Conduct of Research, *Brown University*
2025	Sheridan Teaching Seminar, *Brown University*
2023	Effective Performance, *Brown University*
2022	Carney Computational Cognitive Modeling Workshop, *Brown University*

Service and Outreach

2023-2025	Organizer	The Perception & Action Seminar, *Brown*
2024 & 2025	Booth Presenter	The Brown Brain Fair
2018-2019	Emergency Room Volunteer	The NewYork-Presbyterian Allen Hospital
2017-2019	Volunteer coordinator	KEEN (Kids Enjoy Exercise Now), *New York*

Professional Societies

2023-	Vision Sciences Society
2024 -	International Society for Ecological Psychology

Research Experience

Virtual Environment and Navigation Lab (VENLab) **Aug 2022– Present**
Graduate student researcher - Brown University, Providence, RI

Colwill Lab **Sep 2022– Mar 2023**
Graduate student researcher - Brown University, Providence, RI

The Ro Lab **Aug 2019– May 2022**
Graduate student researcher - CUNY The Graduate Center, New York, NY

Professional Work Experience

Student Disabilities Services - CUNY The Graduate Center **Aug 2019– May 2022**
Graduate assistant

Urbahn Architects **Oct 2016– Dec 2018**
Junior architect and technical writer

Technical Skills

R	Very Good
C#	Good
Python	Good
MATLAB	Fair
Unity3D	Very Good
AutoCAD	Good
Rhino3D	Good